

01-jj-entorno-y-carga

May 14, 2026

1 Proyecto 3 - Entorno y carga de datos

1.1 Integrantes

- Daniel Fernando Salgado Santamaría
- Jairo Wladimir Jhayya Perlaza
- Luis Gabriel Salgado Santamaría
- Oscar Paul Naranjo Castro

Fecha: 2026-05-12

Notebook: 01-jj-entorno-y-carga.ipynb

1.2 Objetivo

Configurar el entorno de trabajo del proyecto, definir rutas reproducibles, cargar los archivos fuente y realizar una validación inicial de estructura, dimensiones y tipos de datos.

1.3 Archivos esperados

- customers.csv
- sessions.csv
- events.csv
- orders.csv
- order_items.csv
- products.csv
- reviews.csv

Python ejecutándose desde:

g:\ProyectosPython\proyecto3\.venv\Scripts\python.exe

PROJECT_ROOT: G:\ProyectosPython\proyecto3\desarrollo3

RAW_DIR: G:\ProyectosPython\proyecto3\desarrollo3\data\raw

FIGURES_DIR: G:\ProyectosPython\proyecto3\desarrollo3\reports\figures

TABLES_DIR: G:\ProyectosPython\proyecto3\desarrollo3\reports\tables

1.4 Verificación de archivos de entrada

En esta sección se comprueba si los archivos fuente existen en `data/raw/` antes de cargarlos.

```
dataset      filename  exists
full_path
```

```

0    customers    customers.csv    True
G:\ProyectosPython\proyecto3\desarrollo3\data\...
1    sessions    sessions.csv    True
G:\ProyectosPython\proyecto3\desarrollo3\data\...
2    events      events.csv    True
G:\ProyectosPython\proyecto3\desarrollo3\data\...
3    orders      orders.csv    True
G:\ProyectosPython\proyecto3\desarrollo3\data\...
4    order_items order_items.csv    True
G:\ProyectosPython\proyecto3\desarrollo3\data\...
5    products    products.csv    True
G:\ProyectosPython\proyecto3\desarrollo3\data\...
6    reviews     reviews.csv    True
G:\ProyectosPython\proyecto3\desarrollo3\data\...

```

1.5 Resumen inicial de dimensiones

	dataset	rows	columns	memory_mb
2	events	760958	10	156.19
1	sessions	120000	6	28.10
4	order_items	59163	5	2.26
3	orders	33580	10	10.34
0	customers	20000	7	4.96
6	reviews	10780	6	1.80
5	products	1197	6	0.18

1.6 Muestra inicial de los datasets

Se visualiza una muestra de cada tabla para verificar estructura, nombres de columnas y contenido general.

	customer_id	name	email	country	age
signup_date	marketing_opt_in				
0	1	Jennifer Salinas	nicholas59@example.org	JP	71
2020-09-04		True			
1	2	Phillip Ramos	christinarubio@example.com	IN	26
2020-04-05		False			
2	3	Dawn Fowler	jessica03@example.org	BR	21
2023-08-31		True			
3	4	Mario Butler	paula27@example.org	FR	63
2022-06-30		True			
4	5	Amber Brown	kevin85@example.net	BR	19
2022-07-22		True			

	session_id	customer_id	start_time	device	source	country
0	1	12360	2021-12-27T00:01:36	mobile	email	DE
1	2	13917	2025-01-31T21:29:42	desktop	organic	PL

2	3	1022	2024-02-19T00:52:50	tablet	organic	FR
3	4	2882	2024-08-04T19:54:31	mobile	direct	GB
4	5	1286	2022-06-28T13:58:08	desktop	email	ES

event_id	session_id		timestamp	event_type	product_id	qty
cart_size	payment	discount_pct	\			
0	1	1	2021-12-27T00:08:36	page_view	93.0	NaN
NaN	NaN	NaN				
1	2	1	2021-12-27T00:16:36	page_view	1005.0	NaN
NaN	NaN	NaN				
2	3	1	2021-12-27T00:18:01	add_to_cart	1005.0	1.0
NaN	NaN	NaN				
3	4	1	2021-12-27T00:45:36	page_view	918.0	NaN
NaN	NaN	NaN				
4	5	1	2021-12-27T01:03:36	page_view	946.0	NaN
NaN	NaN	NaN				

	amount_usd
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

order_id	customer_id		order_time	payment_method	discount_pct
subtotal_usd	total_usd	country	device	\	
0	1	13917	2025-01-31T23:07:42	card	20
107.15	85.72	PL	desktop		
1	2	1022	2024-02-19T01:17:50	card	0
116.17	116.17	FR	tablet		
2	3	6145	2024-12-04T20:24:13	card	0
137.35	137.35	US	mobile		
3	4	3152	2024-07-17T08:50:47	card	15
32.18	27.35	BR	mobile		
4	5	12378	2020-08-21T16:54:16	card	0
238.09	238.09	NL	desktop		

	source
0	organic
1	organic
2	organic
3	email
4	paid

order_id	product_id	unit_price_usd	quantity	line_total_usd
0	1	226	107.15	1
1	2	771	116.17	1

2	3	415	94.49	1	94.49
3	3	24	42.86	1	42.86
4	4	1157	32.18	1	32.18

	product_id	category	name	price_usd	cost_usd	margin_usd
0	1	Electronics	SSD MediumBlue	149	570.28	352.69
217.59						
1	2	Electronics	Keyboard DeepPink	696	498.13	263.13
235.00						
2	3	Electronics	Headphones Orchid	188	548.53	309.60
238.93						
3	4	Electronics	Smartwatch BurlyWood	664	268.36	153.56
114.80						
4	5	Electronics	Smartwatch Cornsilk	328	63.69	42.65
21.04						

	review_id	order_id	product_id	rating	review_text	review_time
0	1	4	1157	2	Not great quality.	2022-02-01T00:00:00
1	2	7	405	4	Good value for money!	2020-03-07T00:00:00
2	3	7	487	5	Excellent product, highly recommend!	2024-12-26T00:00:00
3	4	7	442	5	Excellent product, highly recommend!	2021-03-18T00:00:00
4	5	7	348	4	Good value for money!	2025-09-30T00:00:00

1.7 Tipos de datos y valores nulos

```
=====
CUSTOMERS - INFO
=====
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customer_id           20000 non-null  int64
1   name                  20000 non-null  str
2   email                 20000 non-null  str
3   country               20000 non-null  str
4   age                   20000 non-null  int64
5   signup_date           20000 non-null  str
6   marketing_opt_in      20000 non-null  bool
dtypes: bool(1), int64(2), str(4)
```

memory usage: 957.2 KB

```
=====
SESSIONS - INFO
=====
```

```
<class 'pandas.DataFrame'>
RangeIndex: 120000 entries, 0 to 119999
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   session_id      120000 non-null  int64
1   customer_id     120000 non-null  int64
2   start_time      120000 non-null  str
3   device          120000 non-null  str
4   source          120000 non-null  str
5   country         120000 non-null  str
dtypes: int64(2), str(4)
memory usage: 5.5 MB
```

```
=====
EVENTS - INFO
=====
```

```
<class 'pandas.DataFrame'>
RangeIndex: 760958 entries, 0 to 760957
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  -
0   event_id        760958 non-null  int64
1   session_id      760958 non-null  int64
2   timestamp       760958 non-null  str
3   event_type      760958 non-null  str
4   product_id      682469 non-null  float64
5   qty             143126 non-null  float64
6   cart_size       44909 non-null   float64
7   payment         33580 non-null   str
8   discount_pct    33580 non-null   float64
9   amount_usd      33580 non-null   float64
dtypes: float64(5), int64(2), str(3)
memory usage: 58.1 MB
```

```
=====
ORDERS - INFO
=====
```

```
<class 'pandas.DataFrame'>
RangeIndex: 33580 entries, 0 to 33579
```

Data columns (total 10 columns):

#	Column	Non-Null Count	Dtype
0	order_id	33580 non-null	int64
1	customer_id	33580 non-null	int64
2	order_time	33580 non-null	str
3	payment_method	33580 non-null	str
4	discount_pct	33580 non-null	int64
5	subtotal_usd	33580 non-null	float64
6	total_usd	33580 non-null	float64
7	country	33580 non-null	str
8	device	33580 non-null	str
9	source	33580 non-null	str

dtypes: float64(2), int64(3), str(5)

memory usage: 2.6 MB

=====

ORDER_ITEMS - INFO

=====

<class 'pandas.DataFrame'>

RangeIndex: 59163 entries, 0 to 59162

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	order_id	59163 non-null	int64
1	product_id	59163 non-null	int64
2	unit_price_usd	59163 non-null	float64
3	quantity	59163 non-null	int64
4	line_total_usd	59163 non-null	float64

dtypes: float64(2), int64(3)

memory usage: 2.3 MB

=====

PRODUCTS - INFO

=====

<class 'pandas.DataFrame'>

RangeIndex: 1197 entries, 0 to 1196

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
0	product_id	1197 non-null	int64
1	category	1197 non-null	str
2	name	1197 non-null	str
3	price_usd	1197 non-null	float64
4	cost_usd	1197 non-null	float64
5	margin_usd	1197 non-null	float64

dtypes: float64(3), int64(1), str(2)
memory usage: 56.2 KB

=====

REVIEWS - INFO

=====

```
<class 'pandas.DataFrame'>  
RangeIndex: 10780 entries, 0 to 10779  
Data columns (total 6 columns):  
#   Column          Non-Null Count  Dtype  
---  ----  
0   review_id       10780 non-null  int64  
1   order_id        10780 non-null  int64  
2   product_id      10780 non-null  int64  
3   rating          10780 non-null  int64  
4   review_text     10780 non-null  str  
5   review_time     10780 non-null  str  
dtypes: int64(4), str(2)  
memory usage: 505.4 KB
```

	column	nulls	dataset	null_pct
20	payment	727378	events	95.59
21	discount_pct	727378	events	95.59
22	amount_usd	727378	events	95.59
19	cart_size	716049	events	94.10
18	qty	617832	events	81.19
17	product_id	78489	events	10.31
0	customer_id	0	customers	0.00
1	name	0	customers	0.00
2	email	0	customers	0.00
3	country	0	customers	0.00
4	age	0	customers	0.00
5	signup_date	0	customers	0.00
6	marketing_opt_in	0	customers	0.00
7	session_id	0	sessions	0.00
8	customer_id	0	sessions	0.00
9	start_time	0	sessions	0.00
10	device	0	sessions	0.00
11	source	0	sessions	0.00
12	country	0	sessions	0.00
13	event_id	0	events	0.00
14	session_id	0	events	0.00
15	timestamp	0	events	0.00
16	event_type	0	events	0.00

23	order_id	0	orders	0.00
24	customer_id	0	orders	0.00
25	order_time	0	orders	0.00
26	payment_method	0	orders	0.00
27	discount_pct	0	orders	0.00
28	subtotal_usd	0	orders	0.00
29	total_usd	0	orders	0.00

Archivos guardados en:

G:\ProyectosPython\proyecto3\desarrollo3\reports\tables\dataset_summary.csv

G:\ProyectosPython\proyecto3\desarrollo3\reports\tables\null_summary.csv

1.8 Resultado de esta etapa

En este notebook se dejó configurado el entorno del proyecto, se definieron rutas reproducibles, se verificó la existencia de los archivos fuente y se realizó una carga inicial de todos los datasets.

1.9 Siguiendo paso

El siguiente notebook (02-jj-eda-clientes.ipynb) se enfocará en el análisis exploratorio, calidad de datos y relaciones iniciales entre tablas.